

**Intrusive and Voluntary Memory from a Single System: A Retrieved-Context Account of
Selective Interference**

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Abstract

Visuospatial tasks performed after trauma-film encoding, or after a later reminder of the film, can reduce intrusive memories while leaving voluntary memory relatively intact. This selective interference effect is often treated as evidence that intrusive and voluntary remembering depend on differently vulnerable memory representations. Here we simulate an alternative retrieved-context account using an emotional Context Maintenance and Retrieval (eCMR) model, asking whether context reinstatement, competitor learning, and retrieval control can generate the selective-interference pattern without separately vulnerable representations. In the model, film context remains active after encoding and can later be reinstated by reminders, so an interfering task performed in either state binds task representations to overlapping context. When later retrieval contexts cue both film representations and task competitors, uncontrolled retrieval from context cues is less likely to sample trauma-film items in intrusion-like retrieval, even though film-item strength is not reduced. Deliberate recall is less vulnerable when task goals and control processes reinstate film context and prioritize film items over competitors for retrieval; film-specific cues can also steer retrieval toward film context across test formats. Together, the simulations reproduce selective impairment of intrusion-like retrieval and specify when the intrusion-voluntary-memory dissociation should be more or less pronounced, helping to interpret heterogeneity across empirical paradigms. Selective interference is therefore compatible with a single retrieved-context system and is not by itself diagnostic of separate representational systems or selective trace weakening.

Keywords: intrusive memories, voluntary memory, trauma-film paradigm, visuospatial intervention, selective interference, retrieved-context theory, computational modeling

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Introduction

Traumatic experience foregrounds a tension between durability and selectivity in episodic memory. Highly consequential events should be retained well enough in memory that critical details remain accessible long into the future. At the same time, adaptive retrieval is selective: what comes to mind should normally depend on current cues, goals, and safety. Intrusive memories after trauma suggest that durable encoding can outpace regulatory control. The event returns without being deliberately sought and without being well matched to the demands of the present ([Brewin et al., 2010](#); [Ehlers & Clark, 2000](#); [Iyadurai et al., 2019](#); [Krans et al., 2009](#)). Understanding why such memories arise, and how they differ from voluntary memory for the same event, is therefore a shared target for both clinical science and basic memory theory.

The trauma-film paradigm provides a controlled setting for comparing intrusive and voluntary memory for the same event ([Holmes & Bourne, 2008](#); [James et al., 2016](#)). Participants view distressing film material, record later intrusive memories of the film, and often complete voluntary-memory tests for the same material. This design makes it possible to ask whether the same encoded event can remain available to deliberate retrieval while becoming more or less likely to return involuntarily. In such research, voluntary memory can be relatively preserved, and its relationship to intrusion frequency can be weak, such that the two measures dissociate ([Holmes et al., 2004](#); [James et al., 2016](#)). The same encoded episode can therefore support both uncontrolled and controlled access, and these modes can come apart empirically ([Holmes et al., 2004](#); [Iyadurai et al., 2019](#)).

Visuospatial interference tasks make this dissociation experimentally sharper. When brief tasks such as Tetris are performed after film encoding or after a later film reminder, subsequent intrusions can be reduced while voluntary memory for the same material shows little measurable impairment ([Asselbergs et al., 2023](#); [Deepprose et al., 2012](#); [Holmes et al., 2009, 2010](#); [James et al., 2015](#); [Kessler et al., 2020](#); [Lau-Zhu et al., 2019](#)). Even for memories of real-life traumatic

events, including motor vehicle accidents, emergency caesarean section, and war-related trauma, the same post-reminder intervention can reduce intrusions while leaving deliberate recall relatively intact ([Horsch et al., 2017](#); [Iyadurai et al., 2018](#); [Kanstrup et al., 2021](#)). At the same time, the evidence base is heterogeneous, and recent reviews and replications caution against treating the effect as settled or uniform ([Asselbergs et al., 2023](#); [Varma et al., 2024](#); [Wessel et al., 2025](#)). This selective interference effect therefore raises a theoretical question: when and why can a post-encoding or post-reminder manipulation alter one mode of retrieval more than another, even while both draw on the same recent experience?

One prominent interpretation treats the selective interference effect as evidence for separable memory representations ([Brewin et al., 1996](#); [Brewin, 2014](#)). Under this dual-representation account, traumatic experience produces distinct sensory and contextual memory traces — the first supporting involuntary re-experiencing, the second supporting deliberate recall ([Brewin et al., 2010](#); [Brewin, 2014](#)). A visuospatial interference task competes for the perceptual processing resources needed to consolidate the sensory trace, reducing later intrusions while leaving the contextual trace and the voluntary memory it supports relatively intact ([Holmes et al., 2009](#); [James et al., 2015](#)). When the intervention follows a reminder at longer delays, the account invokes reconsolidation: the reminder reactivates the sensory trace, and the interfering task disrupts its restabilization during a time-limited window ([James et al., 2015](#); [Kindt et al., 2009](#)). The account thus ties selective interference to separate memory representations, modality-specific disruption of one trace, and a discrete reconsolidation window.

Here we argue that selective interference need not be interpreted as evidence for differently vulnerable memory representations. It may instead reflect differences in retrieval processes operating over a shared set of episodic representations. Mechanisms supporting selective, goal-directed retrieval in voluntary recall may be sufficient to sidestep or overcome interference that nonetheless reduces the likelihood of unintentional recall. On this interpretation, the selective-interference effect reflects a capacity to intentionally focus retrieval on goal-relevant event information while reducing the influence of competing memories.

We formalize and evaluate this proposal within a retrieved-context framework drawing on the emotional Context Maintenance and Retrieval model family, including eCMR and CMR3 (Cohen & Kahana, 2022; Talmi et al., 2019). Retrieved-context models explain recall by treating context as a gradually changing state that is updated by studied items, reinstated by recalled items, and used to cue later memory search (Howard & Kahana, 2002; Polyn et al., 2009). Emotional extensions adds source-like emotional context and emotion-dependent binding strength, allowing the same architecture to capture how emotional content changes the organization and success of recall Cohen & Kahana (2022). Variants also model strategic control over memory search, allowing the model to capture how retrieval goals and task demands to shape recall (Cohen & Kahana, 2022; Lohnas et al., 2015; Sederberg et al., 2008). Together, these commitments make the framework a principled setting for testing whether selective interference can arise within a single emotional episodic-memory system.

Consolidation and reconsolidation accounts locate selective interference in a change to the availability of the film representation after encoding or reminder reactivation. A context-binding account shifts the explanatory focus to the retrieval consequences of encoding later experiences in contexts that overlap with the original event (Yonelinas et al., 2019). For selective interference, our model treats the original event, any reminder, and subsequent interference as episodes encoded by binding item representations to a continuously evolving context representation (Howard & Kahana, 2002; Polyn et al., 2009; Yonelinas et al., 2019). Later retrieval is driven by the current context state as a cue, so competition depends on how strongly different episodes are associated with overlapping regions of context. Deliberate recall is more resistant because it combines a more target-oriented initial cue with a sharper decision rule, both of which favor film items over competitors. The dissociation between involuntary and voluntary recall is thus attributed to differences in retrieval state applied to the same stored episodes.

Empirical and Theoretical Target

A Retrieved-Context Account of Selective Interference

Model and Paradigm

Simulation Overview

Results

Simulation 1: Selective Interference Across Retrieval Configurations

Simulation 2: Competitor Encoding in Reinstated Context

Simulation 3: Contextual Overlap Versus Generic Task Load

Simulation 4: Test-Format Heterogeneity

Robustness and Diagnostic Checks

General Discussion

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